

Vignesh S

Software Engineer

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PROFESSIONAL SUMMARY

Technology-driven Software Engineer with nearly 2 years of experience translating business requirements into full-stack solutions — from React/Angular interfaces to scalable backend architecture. Engages across the full development lifecycle, from design through deployment, and drives adoption of new tools and patterns to improve system reliability and performance.

EXPERIENCE

Software Engineer

Mar 2025 – Present

AAPGS Private Limited

Madurai, Tamilnadu

- Optimized CI/CD pipeline by removing redundant steps and adding fail-fast error handling on mid-pipeline failures; diagnosed a recurring heap-memory build failure, cutting deploy time by 70%.
- Reduced database write operations by 98% by buffering IoT payloads via BullMQ-backed Redis queue and flushing via batched writes; migrated from a raw Redis BLPOP list to BullMQ for retry handling, with a day-partitioned recovery log capturing unprocessed data on Redis failures.
- Designed multi-tenant architecture with dynamic per-tenant database switching and in-memory connection caching with 10-minute TTL eviction, reducing MongoDB connection latency by 99% on repeated tenant access while keeping each customer's data isolated on shared infrastructure.
- Built idempotent PayU payment integration with hash-based signature verification and retry-safe handling, and OTP-based delivery verification for UPI orders (validated server-side against PayU payment status before delivery confirmation), preventing duplicate transactions and fraudulent order confirmations.

Web Developer

Dec 2024 – Feb 2025

E-Careerpluz Private Limited

Madurai, Tamilnadu

- Built a responsive business website with a PHP backend handling contact form submissions, subscription emails, and automated notifications.
- Crafted interactive UI using HTML, CSS, and JavaScript for a digital services and legal advisory platform.

PROJECTS

DeployDock | *Next.js, Node.js, TypeScript, MongoDB, SSH, PM2, SSE, Docker* | [GitHub](#)

Personal Project

- Built a self-hosted Continuous Integration/Continuous Deployment (CI/CD) platform with SSH-based remote deployments, PM2 process management, and git-based rollback support.
- Automated deployments via GitHub webhook integration, supporting both SSH-based and Docker-based deployment targets while tracking commit SHAs and full deployment history.
- Streamed deploy logs in real time via SSE backed by worker threads, keeping the deployment pipeline non-blocking under concurrent jobs.
- Built a post-deploy notification system sending deployment status updates via email and WhatsApp, keeping teams informed without manually checking the pipeline.

ProfluxIoT | *Angular, Node.js, MongoDB, Redis, Socket.IO, ECharts*

AAPGS Private Limited

- Architected a scalable IoT platform with per-tenant database isolation and JWT-based authentication.
- Eliminated per-emit MongoDB reads by caching dashboard metadata in Redis with NX locks, reducing query latency on every real-time data push.
- Delivered a configurable dashboard system with drag-and-drop chart and table widgets, supporting multiple dashboards per user each subscribing to its own room for targeted real-time updates.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, C++

Frameworks: React, Next.js, Angular, Node.js, Express.js, Fastify

Databases: MongoDB, Redis, MySQL

Tools & Technologies: Git, GitHub, Docker, Webhooks, Socket.IO, BullMQ, PM2, S3-Compatible Storage

Concepts: JWT, OAuth, REST APIs, CI/CD, Microservices

EDUCATION

The Madura College

Madurai, Tamilnadu

Bachelor of Science in Information Technology, CGPA: 7.61

Aug 2021 – May 2024